Spiro N. Klimentos

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Executive Summary

Software Engineer with 3+ years of experience in C++/Python, backend automation, SQL data workflows, CI/CD, AWS, Docker, and low-latency systems. Proven leader who can both deliver technical work hands-on and manage agile workflows, project timelines, cross-team dependencies, and engineering execution.

Skills

C++, Python, Java, WindRiver Vx-Works, AWS, Git, Docker, Kafka, SQL, Matlab, Simulink, LabView, Project Management, Atlassian Suite, AWS, Git, Docker, Kubernetes, Jenkins, Deep Learning, Controls, Agile/Scrum

Education

Rutgers University - M.S. Aerospace Engineering **–** *New Brunswick, New Jersey* ***May 2025***

- ***GPA: 3.92***

Rutgers University - B.S. Aerospace Engineering **–** *New Brunswick, New Jersey* ***May 2023***

- ***GPA: 3.57***

Professional Experience

L3 Harris Technologies – *Clifton, NJ*

Senior Software Engineer ***Feb 2026 – Present***

- Product Owner and Technical Lead for a team of six software engineers, planning technical milestones, tracking intra/inter team dependencies, and coordinating deliverables across teams.
- Architected a modular C++ solution for software functionality across multiple hardware platforms using driver templatization, factory pattern selection, and adapter interfaces for existing code.
- Enhanced multi-threading scheduler to minimize processing by moving functionality from software to firmware level, and reduced memory leaks; leading to a 30% reduction in CPU usage.
- Refined technical tickets, clarified acceptance criteria, assigned work, and managed sprint execution

L3 Harris Technologies – *Clifton, NJ*

Software Engineer ***Jul 2023 – Feb 2026***

- Designed asynchronous service communication workflows using message-driven APIs, supporting decoupled inter-service messaging, parallel task execution, and priority-based queue processing.
- Developed low-latency predictive algorithms for RF communications in air & space domain systems.
- Built automated data-ingestion workflows for board status data into SQL databases.
- Upkept and deployed Jenkins CI/CD test pipelines to automate validation workflows, improve repeatability, and support faster integration across engineering teams.

Graduate Research Publication – *Piscataway, NJ* ***Aug 2024 – May 2025***

Analysis of Spike Clustering in MEA Data for Neuronal Response Prediction

- Developed a Python-based ML pipeline with TensorFlow and PyTorch to cluster ~1B points.
- Applied dimensionality reduction (PCA, UMAP) and implemented unsupervised clustering using K-Means and Gaussian Mixture Models.
- Standardized data input templates and automated preprocessing through a GUI tool.

Researcher & Co-Author– *Piscataway, NJ* ***May 2022 – May 2024***

Vibrational Response of Boron Nitride Nanofiller in Epoxy Nanocomposites

- Implemented Fast-Fourier Transforms in Python to filter out first-degree harmonic frequencies and Co-authored published paper: DOI: 10.1115/IMECE2024-145125).
- Automated data collection from equipment, improving efficiency and reproducibility of experiments.

AIAA Rutgers Rocket Propulsion Lab – *Piscataway, NJ*

Vice-President & Principal Propulsion Engineer ***June 2020 – Aug 2023***

- Directed a 90+ member team in propulsion R&D, launch logistics, and system safety; won 2nd place in design at international competition, SpacePort America Cup.
- Developed a drone payload mechanism programmed for automated deployment at rocket apogee.
- Created live data collection software in Python to calculate rocket motor performance.

Technical Projects

AI Meal Plan Service – Independent Project ***Sept 2023 – June 2024***

- Built a subscription-based AI meal-planning service using Python backend automation, OpenAI API workflows, and RAG-style retrieval of structured SQL user-profile data.
- Designed backend pipeline for real-time population of user data using Python and Wix integrations.
- Implemented scheduled cloud-based generation workflows for recurring customer deliverables
- Deployed the product to 50 active users providing production support for customer-facing reliability.

Reaction Control System Rocket Stabilization ***Mar 2022 – May 2023***

- Coded PID loop in C++ for rocket to actuate fins to reduce the effects of pitch and yaw.
- Created custom library for real time Kalman Filtering giving accurate telemetry data for rocket.
- Simulated in Ansys Fluent the aerodynamics performances of different fins to find optimal geometry.

NASA RASC-AL Mars Oxygen-Methane Refinement System ***Oct 2021 – Mar 2022***

- Researched and designed a Fission Surface Power system for powering the ISRU operations.
- Proposed a swarm-based rover system for continuous data collection and retrieval of ice.
- Planned power budget and timesheet for all sub-systems to meet 80 kW limit.